Simple & Multiple Regression: OLS and Gauss–Markov

Applied Linear Models I • Lecture 3 of 16

Applied Linear Models I

Today

- Derive OLS from the **normal equations**.
- Sampling properties: unbiasedness and $\mathrm{Var}(\hat{\boldsymbol{\beta}}) = \sigma^2(\mathrm{X}^\top \mathrm{X})^{-1}$.
- The **Gauss–Markov theorem**: OLS is BLUE.
- What the theorem needs – and what it does *not* (no normality).

The OLS estimator

Minimize $S(\mathsf{b}) = \|\mathsf{y} - \mathsf{X}\mathsf{b}\|^2$. Gradient zero gives the normal equations:

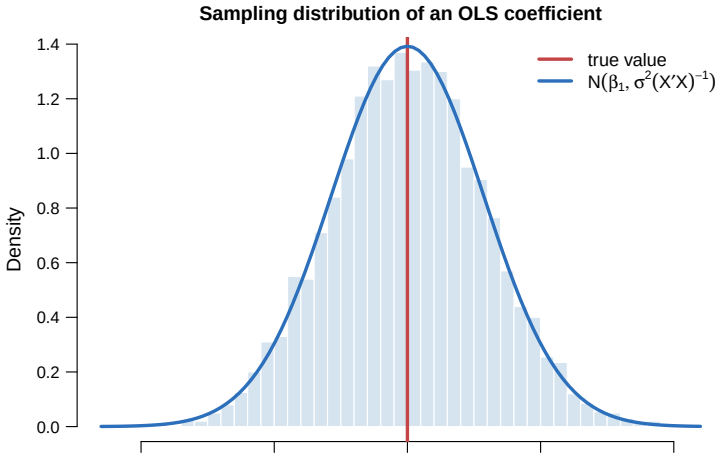
$$\mathsf{X}^\top \mathsf{X}\,\hat{\boldsymbol{\beta}} = \mathsf{X}^\top \mathsf{y} \qquad \Longrightarrow \qquad \hat{\boldsymbol{\beta}} = (\mathsf{X}^\top \mathsf{X})^{-1} \mathsf{X}^\top \mathsf{y}.$$

- Geometrically: the orthogonal projection of y onto $\mathcal{C}(\mathsf{X})$ (Lecture 1).
- In R: `solve(crossprod(X), crossprod(X, y))` matches `lm()` to $\sim 10^{-9}$.
- `lm()` uses QR for numerical stability, but solves the same system.

Unbiasedness

Substitute $y = X\beta + \varepsilon$:

$$\hat{\beta} = \beta + (X^\top X)^{-1} X^\top \varepsilon \quad \Longrightarrow \quad \mathbb{E}[\hat{\beta}] = \beta.$$



Variance of the OLS estimator

Since $\hat{\boldsymbol{\beta}} - \boldsymbol{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \boldsymbol{\varepsilon}$ is linear in $\boldsymbol{\varepsilon}$,

$$\mathrm{Var}(\hat{\boldsymbol{\beta}}) = \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1}, \qquad \hat{\sigma}^2 = \frac{\mathrm{SSE}}{n - p}.$$

- Standard errors $= \sqrt{\mathrm{diag}\left(\hat{\sigma}^2 (\mathbf{X}^\top \mathbf{X})^{-1}\right)}$ – match `lm()` to $\sim 10^{-8}$.
- For predictor j: $\mathrm{Var}(\hat{\beta}_j) = \dfrac{\sigma^2}{(n-1) s_j^2} \cdot \dfrac{1}{1 - R_j^2}$ (the VIF appears here – Lecture 10).

Gauss–Markov theorem

Statement

Under $\mathbb{E}[\varepsilon] = 0$, $\mathrm{Var}(\varepsilon) = \sigma^2 \mathrm{I}$, and X fixed of full rank, OLS has the smallest variance among all linear unbiased estimators. OLS is **BLUE**.

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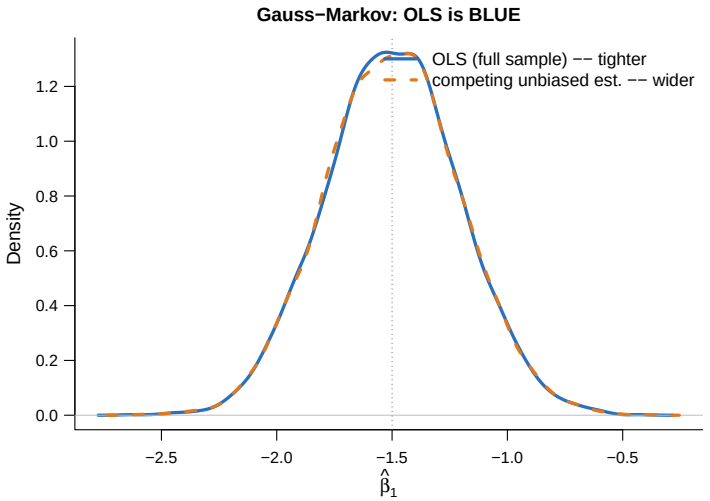
Proof sketch. Any linear estimator $\tilde{\boldsymbol{\beta}} = \mathrm{C}\mathrm{y}$; write $\mathrm{C} = (\mathrm{X}^\top \mathrm{X})^{-1}\mathrm{X}^\top + \mathrm{D}$. Unbiasedness $\Rightarrow \mathrm{D}\mathrm{X} = 0$, so

$$\mathrm{Var}(\tilde{\boldsymbol{\beta}}) = \sigma^2 (\mathrm{X}^\top \mathrm{X})^{-1} + \sigma^2 \mathrm{D}\mathrm{D}^\top \succeq \mathrm{Var}(\hat{\boldsymbol{\beta}}),$$

with equality iff $\mathrm{D} = 0$. ■

Gauss–Markov, empirically

A competing linear unbiased estimator (OLS dropping one row) is unbiased but *wider*:



What Gauss–Markov does *not* require

- **No normality.** Needed only for exact t/F inference (Lecture 4), not for BLUE.
- "Best" is among **linear unbiased** estimators only:
  - non-normal errors $\Rightarrow$ a nonlinear estimator can win;
  - non-spherical errors $\Rightarrow$ GLS is BLUE (Lecture 14);
  - accepting bias can lower MSE (ridge, Stein – Lectures 10, 15).
- Assumes **correct specification** and exogenous X. Omitted variables break unbiasedness itself.

Interpreting a coefficient

In multiple regression $\hat{\beta}_j$ is a **partial effect**: the change in y per unit x_j, *holding all other predictors fixed*.

swiss: Fertility on Education + Catholic

- Holding Catholic fixed: $+1$ Education $\rightarrow -0.79$ Fertility.
- Holding Education fixed: $+1\%$ Catholic $\rightarrow +0.11$ Fertility.

By Frisch–Waugh–Lovell this equals the simple slope on *residualized* variables – "controlling for" = "residualizing out."

Summary

- $\hat{\boldsymbol{\beta}} = (\mathrm{X}^\top \mathrm{X})^{-1} \mathrm{X}^\top \mathrm{y}$; unbiased with $\mathrm{Var} = \sigma^2 (\mathrm{X}^\top \mathrm{X})^{-1}$.
- Gauss–Markov: OLS is BLUE, *without* normality.
- A coefficient is a partial, ceteris-paribus effect.
- Limits of the theorem $\Rightarrow$ the rest of the course.

Next

Lecture 4: inference – t-tests, F-tests, and confidence intervals.